



Figure 1: Grub2 default welcome screen

Grub2: An in-depth look

Grub2 offers new configuration files and is not at all similar to the older version even in terms of how it looks. There are also major changes in the functionality of the new Grub. It displays the menu when the *Shift* key is held while, previously, the *Esc* key needed to be used if only a single Linux installation was found.

The configuration panel has got refurbished in the new avatar. The older `menu.lst` file in the `/boot/grub` folder no longer exists and has been replaced by a new `grub.cfg` file. The new config files are dynamic and depend on many modules for data. As a result the `grub.cfg` gets rewritten after the system update. Once the file gets updated, all the changes will be lost. To make static changes, Grub2 has introduced a new user-defined file called `40_custom`, which is located at `/etc/grub.d` director. This file is independent of the modules, and the changes made will remain untouched irrespective of changes made to the `grub.cfg` file.

The noticeable change in Grub2 is perhaps a complete new nomenclature for the partitions. Grub2 now denotes the first partition as 1, unlike Grub, where the first partition was denoted by 0. However, the first hard disk device is still denoted by `hd0`, keeping the older device convention intact.

Directory layout

/boot/grub: Contains most of the configuration files and modules that are needed for the system to boot. The centralised config file `grub.cfg` resides in this folder.

/etc/grub.d: Contains most of the scripts and user defined files through which `grub.cfg` files are created. When the requisite commands are run, files and scripts within this folder are read to amend changes in the main configuration file, i.e., `grub.cfg`

/etc/defaults/grub: Contains the config files for menu settings that are further implemented in the `grub.cfg` file.



Note: It is not advisable to edit the `grub.cfg` file. There are multiple reasons for this, the most important one being if anything goes wrong, the chances of a boot failure are very high. Also, data in the `grub.cfg` file is dynamic and dependent on other config files. So it is advisable to edit the config files in the other directories upon which the `grub.cfg` files rely.

General configuration

Accompanying new config files, Grub2 now offers a completely new command line interface. With new commands used to manage Grub2, things can be cumbersome at times. To make things easier for new users, I will try to cover the general Grub2 commands that will come handy.

Updating Grub: The `update-grub` command lets you update Grub by rewriting `grub.cfg` file reading various system modules and configuration files. The command will ensure and enlist all the scripts and config files in the Grub2 directories.

Switching menu entries: Since it is not recommended to edit the `grub.cfg` file, let's edit the requisite source file in order to make changes. To make a boot entry default, edit `/etc/default/grub`. The Grub file under the `/etc/default` directory has many useful customisation parameters for editing menus.

To make an entry default, change the numerical value of the entry `GRUB_DEFAULT=0` to whichever entry you wish to boot into automatically. Similarly, you can change the time-out seconds in the same file by editing the `GRUB_TIMEOUT=10` value.

Recovering Grub 2: The most important aspect of configuring Grub is recovery. Most of the time, during the installation of some other operating system, Grub could get lost or replaced with some other boot loader. In that case, you have to re-install Grub. In order to do that, a bootable Linux medium with Grub2 as the main boot loader is required. Once booted into the live environment, open a terminal. As the root user, issue the following commands:

```
mount /dev/sdaXY /mnt
grub-install --root-directory=/mnt /dev/hdZ
```

...where `/dev/sdaXY` is the partition in which `/boot` of the system is installed, and `/dev/hdZ` is the hard disk number in which the `/root` partition resides.

Reinstalling Grub will restore and enlist all the entries and settings that you have previously applied. Once Grub is restored, the `update-grub` command will add newer operating system entries from the installed system.

General customisation options: Grub2 offers many customisation options. Most of the set-up files needed to tweak the default look and feel are available under the `/etc/grub.d` directory.

Working with Grub2

Working with Grub2 can be daunting given the number of changes it carries. So you first need to get acquainted with the